

Login Page & Analytics Dashboard

Goal

Build a Login flow plus a Home Dashboard with analytics cards and charts. All data is mocked client-side—and you’ll also prove mastery of testing, accessibility, performance, error-handling, and architecture.

Requirements

1. Project Setup & Architecture
 - a. Scaffold: Latest Next.js + TypeScript (strict mode enabled).
 - b. Folder Structure
 - c. Type Safety: No use of any.
 - d. [Analytics Dashboard Figma Link](#)
2. Auth Flow
 - a. Route: /login
 - b. Form: Email + Password + “Login” button using reusable form components.
 - c. Validation: inline, onBlur, using a schema (e.g. Zod or Yup).
 - d. Mock Logic:
 - i. 500 ms delay.
 - ii. Success only for [admin@example.com](#) / Password1!.
 - iii. On success, return a mock JWT.
 - iv. On failure, throw “Invalid credentials.”
 - e. State Persistence:
 - i. Store token in React Context or Zustand and Secure, HTTP-only cookie via Next.js API route.
 - ii. Middleware (middleware.ts) to guard /dashboard routes at the edge.
 - f. Error Handling:
 - i. Show inline errors.
 - ii. Gracefully handle unexpected exceptions with a global Error Boundary.
3. Layout & Navigation
 - a. Header:
 - i. Links: Home → /dashboard, placeholder nav items.

- ii. “Logout” button: clears state & cookie, redirects to /login.
- b. Protected Routes:
 - i. Any /dashboard access without a valid token must redirect to /login.
- 4. Dashboard Page
 - a. Route: /dashboard
 - b. Data Cards: As shown in design file
 - c. Charts:
 - d. Mock Data: All data should be under data/dashboard.ts.
- 5. Accessibility
- 6. Performance & SEO
- 7. Testing
 - a. Unit Tests: Jest + React Testing Library for key components (forms, cards).
 - b. Coverage: $\geq 80\%$ on lines/functions.
- 8. Documentation & Architecture Decisions
 - a. README File: Setup, run scripts, test commands.
 - b. Where data is mocked.
 - c. Architecture Decision Record
- 9. Code Style & Limits
 - a. Reusable Components: All UI elements must be built as reusable, well-typed components.
 - b. Max Lines: No component file > 300 lines.
- 10. Deployment: Deploy to any service of choice (vercel, netlify, render etc.)

Deliverables

1. GitHub Repo with:
 - a. Scripts to run and test application.
 - b. Clear branch naming and commit history.
2. README & ADR.
3. Deployment URL.

Note

You are encouraged to enhance your quality of delivery by incorporating the use of AI Assisted Programming tools or IDE